MIL-P-43027D

14 September 1976

SUPERSEDING
MIL-P-43027C
21 March 1969

MILITARY SPECIFICATION

PAPER, CHART, MAP AND BOOK, LITHOGRAPHIC FINISH, COATED AND UNCOATED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers three types of lithographic finish papers for use in printing charts and maps, and reflects requirements parallel to the Government Paper Specification Standards of the Joint Committee on Printing.

* 1.2 Classification. The papers covered by this specification shall be of the following types and grades, as specified (see 6.2).

Type I	-	High Wet Strength Lithographic Map (JCP E-20).
Type II	-	Lithographic Finish Map Paper.
Grade A	-	Deleted.
Grade B	-	Chemical Wood Map, Lithograph-Finish (JCP E-40).
Type III	-	Litho Coated Book (JCP A-205).
Type IV	-	Deleted.
Type V	-	Chart Paper.
Grade A	-	Deleted.
Grade B	-	Deleted.

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: US Army Natick Research and Development Command, Natick, MA 01760, by using the self-addressed standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 9310

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SPECIFICATION

MILITARY

MIL-P-3447 - Paper, Map (Except Diazo-Sensitized), Packaging and Packing of.

STANDARD

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Government Paper Specification Standards.

Published by the Joint Committee on Printing Congress of the United States.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

3. REQUIREMENTS

3.1 Standard sample. The color, finish, formation and cleanliness of the paper must conform to the standard sample as adopted by the Joint Committee on Printing and as furnished by the contracting officer.

3.2 Type I. Type I paper shall conform to all of the requirements for E-20, High Wet Strength Lithographic Map Paper set forth in the Government Paper Specification Standards.

3.3 Type II.

* 3.3.1 Type II, grade A. Deleted.

* 3.3.2 Type II, grade B. Type II, grade B paper shall conform to all of the requirements of E-40, Chemical Wood Map, Lithograph-Finish set forth in the Government Paper Specification Standards.

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* 3.4 Type III. Type III paper shall conform to the requirements of A-205 Litho Coated Book set forth in Government Paper Specification Standards.

* 3.5 Type IV. Deleted.

* 3.6 Type V. Deleted.

* 3.6.1 Type V, grade A. Deleted.

* 3.6.2 Type V, grade B. Deleted.

3.7 Form. The map paper shall be furnished in sheet or roll form, as specified (see 6.2).

3.7.1 Rolls. In addition to the requirements specified in the Joint Committee on Printing Specification Standards, rolls shall be furnished on a nonreturnable substantial fiberboard core having an inside diameter of 3 ± 1/16 inch. The core length shall be not less than the width of the roll material nor greater by more than 1/4 inch. The core shall be plugged at both ends with tightly fitted plugs. The roll diameter shall be as specified (see 6.2) with a tolerance of minus 1 inch or plus 1/2 inch. The rolls shall be suitably restrained to prevent unwinding.

3.8 Workmanship. All types and grades of paper covered by this specification shall be clean and free from any dirt, oil spot, sliver, shive, scale, scab, mottle, hole, tear, cut, wrinkle, fold, chafe mark, scuff mark, slime spot, lump, corrugation, dog ear, foreign or undigested matter, and shall be properly trimmed.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of all inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

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* 4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, or qualified in this specification or applicable purchase documents.

4.3 Inspection of the end item.

4.3.1 Examination of the end item. The end item shall be examined for the defects set forth in the applicable subparagraphs at the inspection levels and acceptable quality levels (AQLs) set forth in 4.3.1.9. The lot shall consist of all material of one type, grade, weight and size offered for inspection at one time. Random samples shall be drawn from each lot of the end item for examination of visual, workmanship, dimensional, count, roll construction, color uniformity, and preparation for delivery defects, as applicable. The lot size shall be expressed in units of reams or rolls, as applicable, for examinations in 4.3.1.1 through 4.3.1.6, and in units of shipping containers for examination in 4.3.1.7 and 4.3.1.8.2.

* 4.3.1.1 Examination for visual defects. The sample unit for this examination shall be one sheet taken from each ream in the sample or one yard of material taken from each roll in the sample of the full width of the roll, as applicable. No sample unit shall be taken from the first or last convolution of a roll. Both sides of the sample unit shall be examined.

Examine	Defects
Appearance	Cleanliness, finish or formation (as applicable) not equal to or better than the standard sample. Color not uniform on both sides; not equal to the standard sample. Presence of dirt or oil spots, slivers, shives, scales, shiners, scabs or mottle larger than or more in number than in the standard sample.

4.3.1.2 Examination for defects in workmanship. The sample unit for this examination shall be one sheet taken from each ream in the sample, or one yard of material taken from each roll in the sample of the full width of the roll, as applicable.

Examine	Defects
Workmanship	Edges not trimmed clean; ragged, torn, uneven or crushed edges. Paper curls, does not lie flat, i.e., not flat after reasonable flexing.

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Examine	Defects
Workmanship (cont'd)	Any hole, tear, cut, wrinkle, fold, chafe mark, scuff mark, slime spot, lump, corrugation or dog ear, which will cause damage to the equipment. Presence of lint, fuzz, or any particles which will pick, lift, fluff or pile on the press blanket. Any foreign or undigested matter. Splice within a sheet. Partial sheet.

4.3.1.3 Examination for defects in roll construction. The sample unit for this examination shall be one roll. Examination shall be made by unwinding the roll.

Examine	Defects
Assembly of roll	Not suitably restrained to prevent unwinding. Material not wound evenly and tightly on roll causing soft or uneven edges or telescoping of roll. Material not wound on substantial core; core broken, collapsed, crushed or mutilated. Plugs omitted from one or both ends; do not fit properly.
Unwinding of roll	Paper slips over core when unwinding. Splice comes apart during unwinding. More than 3 splices per roll. Splice not evenly and neatly made; does not cover entire width of material. Splice taped. Splice not flagged with colored marker.
Workmanship	Any hole, tear, cut, crease, wrinkle, fold, chafe mark, scuff mark, slime spot, lump, or corrugation. Edges not trimmed clean; ragged, torn, uneven or crushed edges.

* 4.3.1.4 Examination for defects in dimensions. The sample unit for this examination shall be one sheet taken from each team in the sample or one roll, as applicable.

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Examine	Defects
Sheets	
Sheet size	Less than specified in either direction.
Sheets 8-1/2 by 14 inches or less	Oversize by more than 1/16 inch in either dimension from size specified.
Sheets over 8-1/2 by 14 inches	Oversize by more than 1/8 inch in either dimension from size specified.
Squareness	Dimensional variation exceeds 1/32 inch for sheets 32 inches or less in length, or exceeds 1/16 inch for sheets of over 32 inches in length.
Rolls	
Width	Less than specified. Oversize by more than 1/8 inch.
Diameter	More than one inch less than specified. Over one half inch more than specified.
Core	Length less than width of roll or greater by more than 1/4 inch. Inside diameter less than 2-15/16 inches or greater than 3-1/16 inches.

4.3.1.5 Examination for count (sheets only). The lot shall be unacceptable if the average count per package of quantity between markers, when paper is skid packed, is less than specified.

4.3.1.6 Examination for weight per roll. The lot shall be unacceptable if the average weight per roll including nonreturnable core, plugs and wrapper, for all rolls examined, is less than indicated.

4.3.1.7 Examination for color uniformity. The sample unit for this examination shall be one shipping container or one roll, as applicable.

Examine	Defect
Color uniformity	Color not uniform throughout the shipping container or roll as applicable.

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4.3.1.8 Examination of preparation for delivery.

4.3.1.8.1 Sheets. An examination shall be made to determine that packaging, packing and marking requirements comply with MIL-P-3447 for sheets.

* 4.3.1.8.2 Rolls. An examination shall be made to determine that the packaging, packing and marking requirements comply with MIL-P-3447 for rolls.

* 4.3.1.9 Inspection levels and acceptable quality levels (AQLs) for examinations. The inspection levels, for purposes of determining the sample size, and the acceptable quality levels (AQLs) expressed in defects per 100 units shall be as follows:

Examination paragraph	Inspection level	AQL
4.3.1.1	I	2.5
4.3.1.2	III	0.065
4.3.1.3	S-1	2.5
4.3.1.4	S-2	1.0
4.3.1.5	S-2	N.A.
4.3.1.6	S-2	N.A.
4.3.1.7	S-2	2.5

* 4.3.2 Testing of the end item. The map paper shall be tested for those requirements for each type and grade of paper specified in section 3. The characteristics to be tested for number of determinations per sample unit, test procedures and "results reported as", shall be as specified in Part 2 - Testing Standards of the Government Paper Specification Standards. The lot size for the purpose of determining the sample size shall be expressed in units of reams or rolls of one type and size of map paper. The test sample unit shall consist of a sufficient number of sheets from each ream, or yards of paper from each roll in the sample to provide not less than 1000 square inches of paper. Determinations for moisture content as performed by the contractor shall be made after the paper is trimmed and just prior to wrapping the reams in skid form or on rolls, as applicable. The inspection level, for the purpose of determining the sample size in accordance with MIL-STD-105, shall be S-1 and the acceptable quality level (AQL) shall be 2.5, expressed as defects per one hundred units.

* 5. PACKAGING

* 5.1 Preservation-packaging. Preservation-packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A.

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5.1.1.1 Sheets. Map paper sheets of one type, grade and size only, shall be packaged in accordance with the applicable requirements of MIL-P-3447.

5.1.1.2 Rolls. Packaging not applicable.

5.1.2 Level C.

5.1.2.1 Sheets. Map paper sheets of one type only, shall be packaged in accordance with the applicable requirements of MIL-P-3447.

5.1.2.2 Rolls. Packaging not applicable.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Level A packing.

5.2.1.1 Sheets. Map paper sheets of one type, grade and size only, shall be packed in accordance with the applicable requirements of MIL-P-3447.

* 5.2.1.2 Rolls. Each roll of map paper shall be packed in accordance with the applicable requirements of MIL-P-3447.

5.2.2 Level B packing.

5.2.2.1 Sheets. Map paper sheets of one type, grade and size only, shall be packed in accordance with the applicable requirements of MIL-P-3447.

* 5.2.2.2 Rolls. Each roll of map paper shall be packed in accordance with the applicable requirements of MIL-P-3447.

5.2.3 Level C packing.

5.2.3.1 Sheets. Map paper sheets shall be packed in accordance with the applicable requirements of MIL-P-3447.

* 5.2.3.2 Rolls. Rolls of map paper shall be packed in accordance with the applicable requirements of MIL-P-3447.

5.3 Marking. Marking shall be in accordance with the applicable requirements of MIL-P-3447.

6. NOTES

6.1 Intended use.

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6.1.1 Type I. Type I papers covered by this specification are intended to be used for high quality multi-color letterpress and offset printing of field books, maps, folded inserts, etc., requiring high wet strength and water-resistant properties and good folding endurance. Printed matter may include text, line illustrations, and halftones. Not suitable for halftone production by letterpress printing.

6.1.2 Type II.

* 6.1.2.1 Type II, grade A. Deleted.

6.1.2.2 Type II, grade B. Type II, grade B papers are intended to be used for high quality multicolor letterpress and offset printing of maps, folded inserts, etc., requiring good folding endurance. Printed matter may include test, line illustrations and halftones. Not suitable for halftone production by letterpress printing.

6.1.3 Type III. Type III papers are intended to be used for multicolor face and back offset printing, principally of maps and charts. Printed matter may include text, line illustrations and halftones up to 150-line screen.

* 6.1.4 Type IV. Deleted.

* 6.1.5 Type V. Deleted.

* 6.1.5.1 Type V, grade A. Deleted.

* 6.1.5.2 Type V, grade B. Deleted.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number and date of this specification.
(b) Type, grade, form, and size required (see 1.2 and 3.7).
(c) Roll diameter, when required (see 3.8).
(d) Weight required.
(e) Grain direction.
(f) Pressroom conditions.
(g) Selection of applicable level of packaging and packing (see 5.1 and 5.2).

6.3 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications,

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corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations or relationship to the last previous issue.

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Army - GL

Project No. 9310-0083

Review activity:

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User activities:

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DOCUMENT IDENTIFIER AND TITLE
MIL-P-43027D, Paper, Chart, Map and Book, Lithographic Finish, Coated and Uncoated

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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   ☐ YES ☐ NO *(If "Yes", in what way?)*

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